

Visibility Bias — 1–page summary

Measuring Visibility Bias in Bureaucratic Text: A Validated Instrument with Evidence from Chinese Municipal Government Work Reports

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One–paragraph summary

I introduce the **Visibility Attention Index (VAI)**, a validated text–based instrument for measuring *compositional policy attention* in bureaucratic documents, applied to **6,294 Chinese municipal government work reports** (282 prefectures, 2002–2024). The instrument operationalizes a cadre–attention model in which officials allocate rhetorical emphasis between *visible* (streetscapes, greening, façades) and *functional* (water, drainage, structural safety) urban–infrastructure categories under a principal with asymmetric observability. I subject the instrument to a **five–test construct–validity battery** including, as the key methodological innovation, a **within–document retrospective–vs–prospective differential** ($\Delta = +0.025$, paired $t = 8.4$, $p < 10^{-16}$) that exploits the temporal section structure of bureaucratic documents. Two applications demonstrate utility (compositional substitution in MOHURD accounting data: $\beta = +0.111$, $p = 0.002$; structural welfare calibration yielding ¥1–15B/year). Two pre–registered null results are reported transparently, including one that reverses when the sample is extended to heterogeneity–robust estimation.

Key contribution (methodological)

The **within-document behavioral signature** — showing that retrospective report sections are systematically more visibility-loaded than prospective sections, using the *same lexicon, same author, same report* — is, to my knowledge, novel in the text-as-data literature. It cannot be produced by lexicon-selection bias, author-style effects, or report-length artifacts. It is a diagnostic that any annual bureaucratic-report corpus (EU NRPs, Indian state budgets, Mexican municipal plans) could reproduce.

What I am asking for

I would value **substantive feedback** on: – Whether the construct-validity protocol meets your bar for a measurement instrument in this genre – Whether the domain-scope claim (governance-rhetoric, full stop) is defensible given the Wikipedia-null finding – Whether the within-document differential (E-B) is as novel in the text-methods literature as I believe it is

I am particularly interested in co-authorship on the measurement-focused paper if the construct-validity battery meets your standards. I am finalizing the manuscript and would welcome your engagement at whatever level works for your schedule (feedback only, co-author, or acknowledgment).

Key evidence in 5 figures

1. **E-A Dictionary robustness:** $r(\text{VAI_original_42terms}, \text{VAI_expanded_148terms}) = 0.93$ across 5,686 GWRs
2. **E-B Within-document signature:** review — plan $\Delta = +0.025$, paired $t = 8.4$, $p < 10^{-16}$ ($N = 4,330$)
3. **E-D Cross-source validation:** $r(\text{VAI}, \text{MOHURD yearbook CIR}) = 0.24$, $p < 10^{-30}$
4. **E-F Wikipedia null (scope boundary):** $r(\text{VAI}, \text{Wiki-derived VAI}) = -0.15$, 95% CI $[-0.31, +0.02]$; interpretation: governance-rhetoric vs encyclopedic-description domain mismatch
5. **P1 Accounting-data application:** $\text{CIR} = \beta \cdot \text{VAI} + \text{controls} + \text{FE}$, $\beta = +0.111$, $p = 0.002$ across 2,751 city-years

Pre-registration discipline

All five construct-validity tests pre-registered before execution. Five deviations from pre-registration are logged in Appendix D with reasons, executed substitutes, and effect-on-sign/significance. Two results — H2 inspection event study (failed under heterogeneity-robust extension) and H5 CFPS amenity test (not executable as pre-registered, substitute returned null) — are reported transparently.

What is NOT claimed

This is explicitly not a causal-identification paper. The event-study approach to anti-corruption inspection shock *does not* survive heterogeneity-robust estimation when the window is extended to 2017. I report this prominently rather than bury it.

Full manuscript (~9,500 words) and replication archive available upon request.